

Engineering Notes

28 GHz Microwave-Powered Propulsion Efficiency for Free-Flight Demonstration

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I. Introduction

WIRELESS power transmission (WPT) via microwave has the potential to drastically improve the flight performance of aerospace craft, which require a high-energy-density supply when in flight. WPT technology can provide rockets high specific thrust (low fuel consumption) without the limitations imposed by the chemical energy provided by conventional propellant [1,2]. However, if microwaves dissipate before reaching the rocket, then the overall propulsion efficiency will be very low, regardless of the efficiency of the rocket. In gyrotron experiments, the overall efficiency of WPT for microwave rockets and the total propulsion system efficiency were 14 and 6%, respectively, using a fixed-position rocket [3]. The next step is the demonstration of free flight to determine the total efficiency for the rocket (i.e., including the WPT). However, the flight demonstration requires an over-MW-class continuous-wave microwave source, which is hard to procure and difficult to use in this level of concept study. We therefore demonstrate microwave-powered flight by using unmanned aerial vehicles (UAVs) to evaluate WPT efficiency for aerospace craft [4,5]. WPT technology uses ground-emitted microwaves to charge aerospace craft in flight via a rectenna (rectifier and antenna), which converts microwave power to DC power. To date, WPT studies involving UAVs have mainly used frequencies around 2.45 GHz (industry science and medical band) and were done in the 1980s and 1990s [e.g., the Brown (NASA) demonstration, the SSPS project of Kyoto University, or the Canadian SHARP experiment; see Table 1] [6–9].

Our primary demonstration used a tracking system and involved uniform circular motion of UAVs in a plane. The transmission frequency was 5.8 GHz and the transmission distance was 1500 mm [4]. The results of this demonstration indicate that higher operating

frequencies for microwave and free-flight control systems are necessary to investigate the feasibility of WPT for UAVs. Considering the directivity of microwave beams, a high operating frequency increases the transmission efficiency for UAVs. Assuming a Gaussian beam profile for microwave propagation through free space, the minimum focal-spot diameter is on the order of the microwave wavelength. Conversely, radiation loss from the microwave circuit increases with microwave frequency.

A previous study developed a prototype of a 28 GHz microwave power transfer system for UAVs consisting of an AR.Drone 2.0 (Parrot Inc.), which uses a proportional-integral-differential (PID) algorithm for autonomously controlled systems, and an indoor GPS to direct the high-power microwave radiation in a shielded room [5]. The operating frequency of 28 GHz is the same as in the previous rocket demonstration, and the flight altitude is also the same [3]. According to the flight demonstration with a microwave power source, the maximum power transmission efficiency was about 0.001%, which was done with 800 mm separating the UAV from the microwave transmitter. To enhance the WPT efficiency and evaluate the feasibility of the WPT system, location scattering of UAVs must be avoided; this originates from the communication between the UAVs and the ground computer through Wi-Fi. In the present work, we report the WPT efficiency with a beam-tracking system that synchronizes the UAV position and with the microwave phase tuned to improve the power-transmission efficiency. The microwave phase is controlled by an analog phase shifter synchronized with the indoor GPS. In addition, we present an analytical model for predicting the WPT efficiency of UAVs and compare its results with free-flight and fixed-flight [3] experimental results.

II. Millimeter-Wave Transmission System

A. UAV Control and Tracking System

Figure 1 shows a schematic of the beam-tracking system and UAV position-correction system. It calculates the angle from the UAV pilot signal and gives the phase information to the phase shifter. When the UAV is at angle α as viewed from the antenna, the beam angle is calculated from the path-length difference of the wavefront and the phase difference given to the phase shifter:

$$\alpha = \sin^{-1} \left(\frac{\lambda \Phi}{2\pi d} \right) \quad (1)$$

Here, Φ , λ , and d are the input phase angle, the microwave wavelength, and the distance between transmitters, respectively. The pilot signal is obtained from the indoor GPS units, and tracking is accomplished by constantly updating the calculations based on the UAV movement. The advantage of using beam tracking for the UAV is that it suppresses the reduction in received DC power due to horizontal-movement errors of the UAV and fluctuations in roll and pitch angles. The UAV tracking activates only for the x -axis direction. In the experiment, the UAV hovered at 0.8 m, so there was no need to consider changes in height.

Table 2 lists the characteristics of the AR.Drone 2.0 (Parrot) UAV used in the experiments [10]. The AR.Drone 2.0 optimizes the PID gain by using iterative feedback tuning. Details on the gain optimization are given by Nguyen et al. [5]. The autopilot centerpiece hosts a 32-bit ARM Cortex A8 processor with an 800 MHz video digital-signal processor model TMS320DMC64x. The UAV embedded operating system was Linux 2.6.32 with 1 GB DDR2 RAM running at 200 MHz. The optimization gives a horizontal movement error during hovering of 0.2 ± 0.1 m. Figure 2a shows changes in the transmitter antenna gain with and without tracking by

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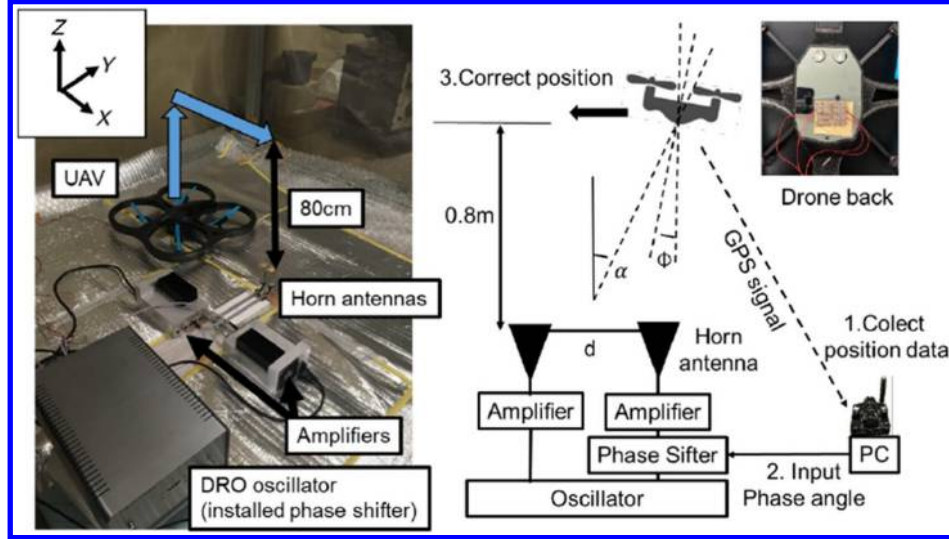
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Table 1 Summary of previous and present free-flight WPT demonstrations

Parameters	Brown (1963)	MILAX (1992)	SHARP (1987)	Present
Operation frequency (GHz)	3.0	2.45	2.45	28
Transmitter power (W)	5,000	1,000	10,000	5
WPT efficiency (%)	15	10	1.5	0.9
WPT distance (m)	5.5	10	150	0.8
Beam control	n/a	Camera based	n/a	GPS based
Type of aircraft	Helicopter	Airplane	Airplane	Drone
Gross weight (kg)	2.3	1.8	4.1	0.42

**Fig. 1** Schematic diagram of UAV microwave transmission and beam-tracking system.

the beam-tracking system. With tracking, the beam half-width is $\pm 10^\circ$, and the beam-tracking system covers 70% of the horizontal error. Conversely, without tracking, the null occurs at approximately $\pm 10^\circ$. Five indoor GPS units (HW v4.9, Marvelmind Robotics) were used to acquire location data for the pilot signal and UAV-position correction. The GPS location-update rate was 7.4 Hz. One GPS unit was mounted on the UAV as a hedge beacon, and the other four were mounted as stationary beacons on pillars in an electromagnetically shielded room.

B. Millimeter-Wave Transmission and Receiver System

The transmitter antennas consist of two 19.2 dBi horn antennas (LB-28-20-C-KF, A-INFO). The cross section of the exit of a single horn antenna is 32 mm $\times$ 40 mm. The horn antenna array is 64 mm $\times$ 40 mm; the longer side is the x direction. Figures 2a and 2b show the power distribution of the 28 GHz electromagnetic waves emitted from the two horn antennas. The beam profile is approximated as a Gaussian beam with $\omega_{0x} = 0.050$ m and $\omega_{0y} = 0.011$ m. At 0.8 m, the beam profile is considered far field. The phase angle is calculated and sent to the analog phase shifter (SKP-0631831036-SFSF-A1, SAGE Millimeter, Inc.) built into the dielectric resonance oscillator (VG14-28G, Vega Technology, Inc.). Finally, beamforming was done towards the UAV. The WPT circuit consisted of the dielectric resonance oscillator, solid-state power amplifier (WPAC172PG, Advanced Microwave, Inc.), isolator, coaxial-WR28 converters, and 19.2 dBi horn antennas. The transmitted power was 5.3 W.

For the receiver side, a 28 GHz rectenna received the microwave power and rectified it to DC power [11]. A DiClad 880 substrate served for both antenna and rectifier (Rogers Corp., relative permittivity = 2.17 and thickness = 0.50 mm). The antenna consisted of a four-element microstrip array with $S_{11, \text{antenna}} = -19.4$ dB and $G_r = 10.7$ dBi at 28 GHz. Figures 2c and 2d show the radiation pattern of the antenna. The measured pattern is consistent with the simulations.

Table 2 Drone specifications (AR.Drone 2.0) [10]

Parameters	Values
Size	52 cm $\times$ 52 cm $\times$ 11 cm
Gross mass	420 g
Battery	11.1 V, 1000 mAh
Flight duration	12 min
Power consumption	55.5 W
Payload	100 g

The 3 dB point of the main lobe appears at -12 to 21° along the E plane and -20 to 10° along the H plane. In the experiment, the E plane corresponds to the roll angle, and the H plane corresponds to the pitch angle. The rectifier circuit consisted of a single shunt type with a class-F load, which reflects higher harmonics that re-enter the GaAs Schottky diode (MA4E1317, MACOM) to increase the RF-DC conversion efficiency. The parameter $S_{11, \text{rectifier}}$ decreases proportionally to the RF power input to the rectifier, as shown in Fig. 2e. Figure 2f shows the RF-DC conversion efficiency as a function of input RF power to the single rectifier terminated at 132 Ω . The maximum value was 0.56 for an input power of 245 mW. When the input exceeds the maximum operating point of 368 mW, the diode breaks down and no longer functions as a rectifier. The single rectenna that integrates the antenna and rectifier circuit is 18 mm $\times$ 15 mm with a mass of 0.47 g.

III. Analytical Formulation of Microwave Power Transmission Efficiency

The analytical formulas verified by the flight experiment can be used to estimate the DC power received in flight with the tracking system. When including all the factors of WPT, the analytical transmission efficiency η_A as a function of operational frequency is

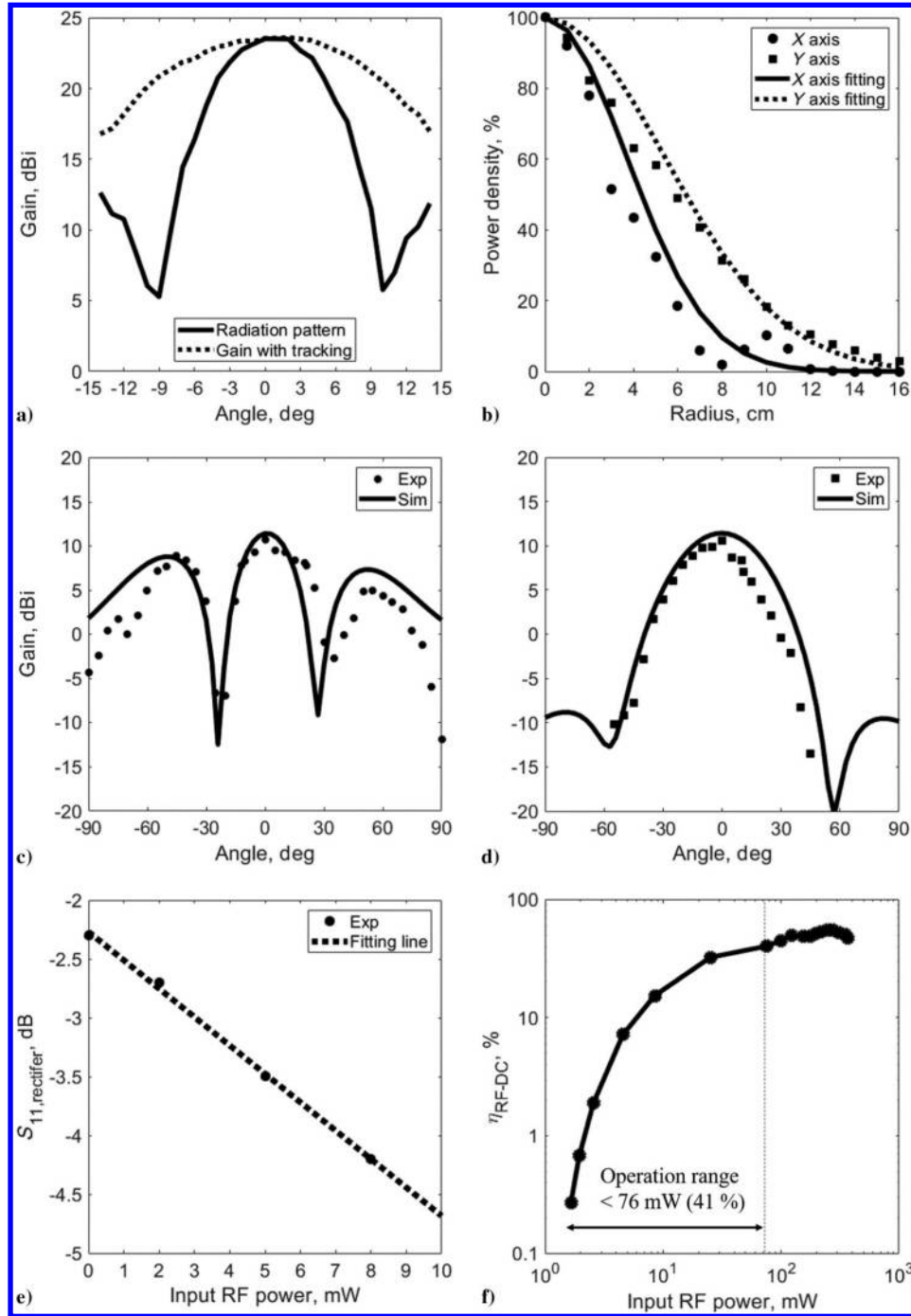


Fig. 2 a, b) Transmitter-side antenna radiation pattern. Rectenna antenna radiation for c) E plane and d) H plane. e) S_{11} parameter on rectenna. f) RF-DC conversion efficiency.

$$\eta_A = \eta_{\text{com}} \eta_{\text{beam}} \eta_{\text{air}} \eta_{\text{cap}} \eta_{\text{tra}} \eta_{\text{RF-DC}} \quad (2)$$

where the subscripts on the right-hand side of Eq. (2) indicate the combined efficiency of the rectenna DC power, the beam efficiency, the efficiency due to atmospheric attenuation, the efficiency of microwave capture, and the RF-DC conversion efficiency, respectively. These efficiencies from the antenna and rectenna characteristics and from ground experiments done independently of flight experiments allow the efficiency of power transmission and reception to be predicted analytically. The combined efficiency $\eta_{\text{com}} = 0.98$ because the combined loss from the DC outputs of all rectifiers is 2% [12]. The beam efficiency η_{beam} is expressed in terms of the radiation pattern of the transmitting antenna and can be approximated in terms of a Gaussian beam:

$$\eta_{\text{beam}} = \left\{ \text{erf} \left(\frac{\sqrt{2}L_x}{2\omega_{zx}} \right) \text{erf} \left(\frac{\sqrt{2}L_y}{2\omega_{zy}} \right) \right\} \exp \left[f - C_{\text{beam}} \alpha^2 \right] \exp \left(-\frac{2y^2}{\omega_{zy}^2} \right) \quad (3)$$

Here, the error function erf is the solution of the Gaussian beam spatial integral; ω and L are the beam radius and the length of rectenna element, respectively; and C_{beam} is the gain coefficient, which decreases from beamforming and is 0.01 based on the fitting parameter of transmitter radiation pattern. The first factor of Eq. (3) is the ratio of the power density for the Gaussian beam as integrated with the rectenna area to the transmitted power. The second factor gives the angle dependence of the transmitter gain for the array

directivity due to beamforming. The last factor describes the decreased collection power as calculated from the horizontal movement of the UAV. In addition, η_{air} for 28 GHz transmission is negligible because of the short distance in the expression for η_A . The efficiency η_{cap} is the ratio of power input into the circuit to power irradiating the rectennas, which is

$$\eta_{\text{cap}} = \eta_{\text{dis}} \frac{\lambda^2 G_{\text{max}}}{4\pi A} f_E(\alpha - \phi) f_H(\theta) \quad (4)$$

where G_{max} , A , θ , and ϕ are the maximum receiver antenna gain, the area of the single receiver antenna, the pitch angle along the H plane, and the roll angle along the E plane, respectively. The distance between each rectenna element is 1.0λ , and Shinohara obtained a 25% reduction in efficiency for a 1.0λ spacing [13]. Thus, the first factor $\eta_{\text{dis}} = 0.75$, and the second factor gives the maximum antenna aperture efficiency. The third and fourth factors are the fitting functions of the receiver antenna gain on the E and H planes, respectively, which serve to estimate the decrease in gain due to the beam and the receiving angle. The quantities α , ϕ , and θ are the input parameters for this analytical formula and are calculated from the flight log of the attitude. Note that, for the sake of simplicity, η_{cap} does not include the effect of the yaw angle. Because the temporal fluctuation is less than $\pm 2^\circ$, the polarization effects on the antenna carry a slight importance in this experiment. The efficiency η_{tra} due to atmospheric transparency of the antenna and the rectifier circuit (see Fig. 2e) is evaluated by using S_{11} as follows:

$$\eta_{\text{tra}} = 1 - 10^{\frac{S_{11,\text{antenna}}}{10}} - 10^{\frac{S_{11,\text{rectifier}}}{10}} \quad (5)$$

Finally, the efficiency $\eta_{\text{RF-DC}}$ is the ratio of the DC power output to the RF power input at the rectifier; $\eta_{\text{RF-DC}}$ is the fitting function of the RF-DC conversion efficiency, as shown in Fig. 2f. According to Eqs. (2–5), with the present transmission system, the total microwave power transmission efficiency for 0.8 m distance is estimated to be 16%. Note that a 300–500 W amplifier is necessary in the ground transmission system to maximize the RF-DC efficiency. In addition, we mount the maximum rectennas possible on the back of the drone.

IV. Demonstration of Millimeter-Wave Power Transmission to UAVs

The flight demonstration experiment was carried out both with and without the beam-tracking system. The UAV was set to hover for 30 s at a height of 0.8 m from the transmitter. Note that the ground effects on the drone can be ignored above a height of 0.78 m [14]. The rectennas were arranged in a 2×2 pattern with parallel connections, and the DC power was detected by using a microcomputer. Figure 3 shows the locations of the rectenna array attached at the body center of the UAV projected onto the $x - y$ plane during the autonomously controlled hovering test at an altitude $h = 800$ mm.

Figure 4a shows the DC output power from the UAV-mounted rectifier for the system with and without the beam-tracking system. A DC output power of 10 mW was obtained with the drone and microwave correctly positioned. With the beam-tracking system, the minimum DC output power was 10^{-2} mW. Conversely, the system without beam tracking could not receive the microwave power in some cases. To determine how beam tracking or drone fluctuations affect the results in Fig. 4a, the angle and the displacement error of the drone are shown in Figs. 4b and 4e, respectively. Figure 4b shows the roll, pitch, and yaw angles, and Fig. 4c shows the drone angle toward the ground antenna. In Figs. 4b and 4c, the roll, pitch, and yaw angles are temporally stable for both cases (i.e., with and without beam tracking). These angles fluctuate less than 3° . Figures 4d and 4e show the $X - Y$ horizontal displacement with error and the drone altitude as a function of time, respectively. The data for they direction show that the drone temporarily flies out of the beam, causing a sudden descent. The flight instability is due to losses in communications with indoor GPS units. In addition, the displacement errors for x , y , and h differ

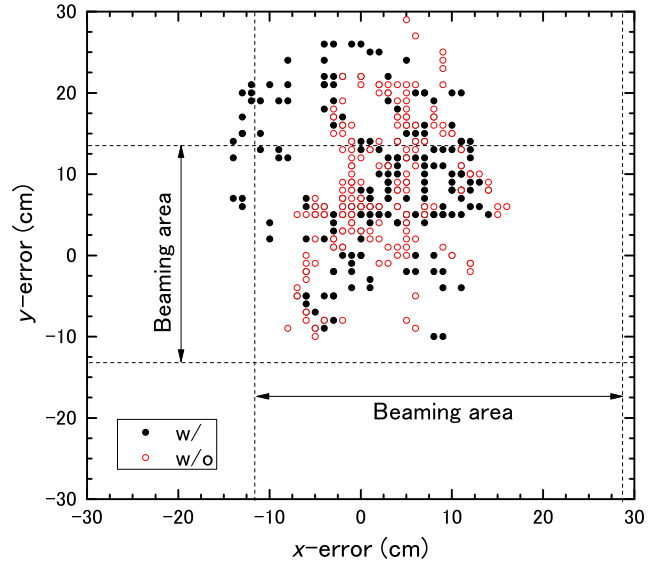


Fig. 3 UAV footprints on $X - Y$ plane during the autonomously controlled hovering test at altitude $d = 800$ mm.

slightly in Figs. 4d and 4e. The error ranges for x and y are about -15 to 15 cm and -10 to 25 cm for beam tracking on and off, respectively. For h , the displacement errors of the system are -40 to 0 cm and -20 to 20 cm with and without beam tracking, respectively. Because the angles α differ slightly in each case, the beam tracking system affects the DC output power in Fig. 4a.

Figure 4f compares η_{expt} with η_A for the system with beam tracking. The maximum η_{expt} is 0.9% at $t = 15.4$ s. η_A using Eq. (2) predicted η_{expt} excepting the some cases. The blue highlighting in Figs. 4b, 4d, 4f, and 4g shows that the drone angle α exceeds 5° , as shown in Fig. 4b. In Fig. 4f, the blue highlighting roughly matches the time when the DC output is less than 10 mW. Note the large gap between η_{expt} and η_A at 11 – 12 s and 26 – 28 s. Explaining these errors in drone position or why the drone left the beam is difficult. One possible reason is that the unexpected reflection of microwave on the ground or the wall might explain the gap between η_{expt} and η_A . In addition, estimating the sidelobe irradiation outside the beamforming range is difficult because η_A can only be applied to the mainlobe. The beam shape in the near field cannot be considered to be the same as that at 0.8 m because the shape is a function of altitude. These problems could be solved in the outdoor demonstration with the long transmission distance.

To analytically understand the experimental results, we break down η_A in the overall flight in Fig. 4g. At $t = 15.4$ s, the maximum η_{expt} and η_A are, respectively, 0.9 and 0.43% . η_A can be broken down as follows: η_{beam} , η_{cap} , η_{tra} , and $\eta_{\text{RF-DC}}$ are 4 , 30 , 90 , and 40% , respectively. The efficiency η_{cap} drops to 25% or less when the beam angle in Fig. 4d exceeds 5° . A 5° swing in the beam angle produces a 7 cm displacement in the UAV position, which reduces η_{cap} to 26% at 0.8 m. $\eta_{\text{RF-DC}}$ and η_{tra} depend on the input power, which depends on the other three efficiencies. When $\alpha > 10^\circ$ ($t = 7$ – 9 s and 19 – 20 s), $\eta_{\text{RF-DC}}$ and η_{tra} decrease drastically to less than 0.001 and 30% , respectively. In addition, the maximum η_{beam} is 5% because the UAV carried only four rectennas and fluctuations in the horizontal motion were 20 cm in this experiment. From Eq. (4), η_{beam} improves with a larger rectenna array and more precise control. When the maximum rectenna array (15 cm $\times$ 9 cm) is loaded on the AR.Drone, η_{beam} is estimated to be 4.2 times more effective than the experimental value at 0.8 m. In addition, if the horizontal error improves to 10 cm, the x and y components of the η_{beam} are 2.6 and 2.1 times more effective, respectively.

The present report thus provides quantitative results showing the difference between fixed-flight and free-flight transmission efficiency. For the free-flight demonstration, the microwave tracking system allowed stable power beaming to the drone most of the time.

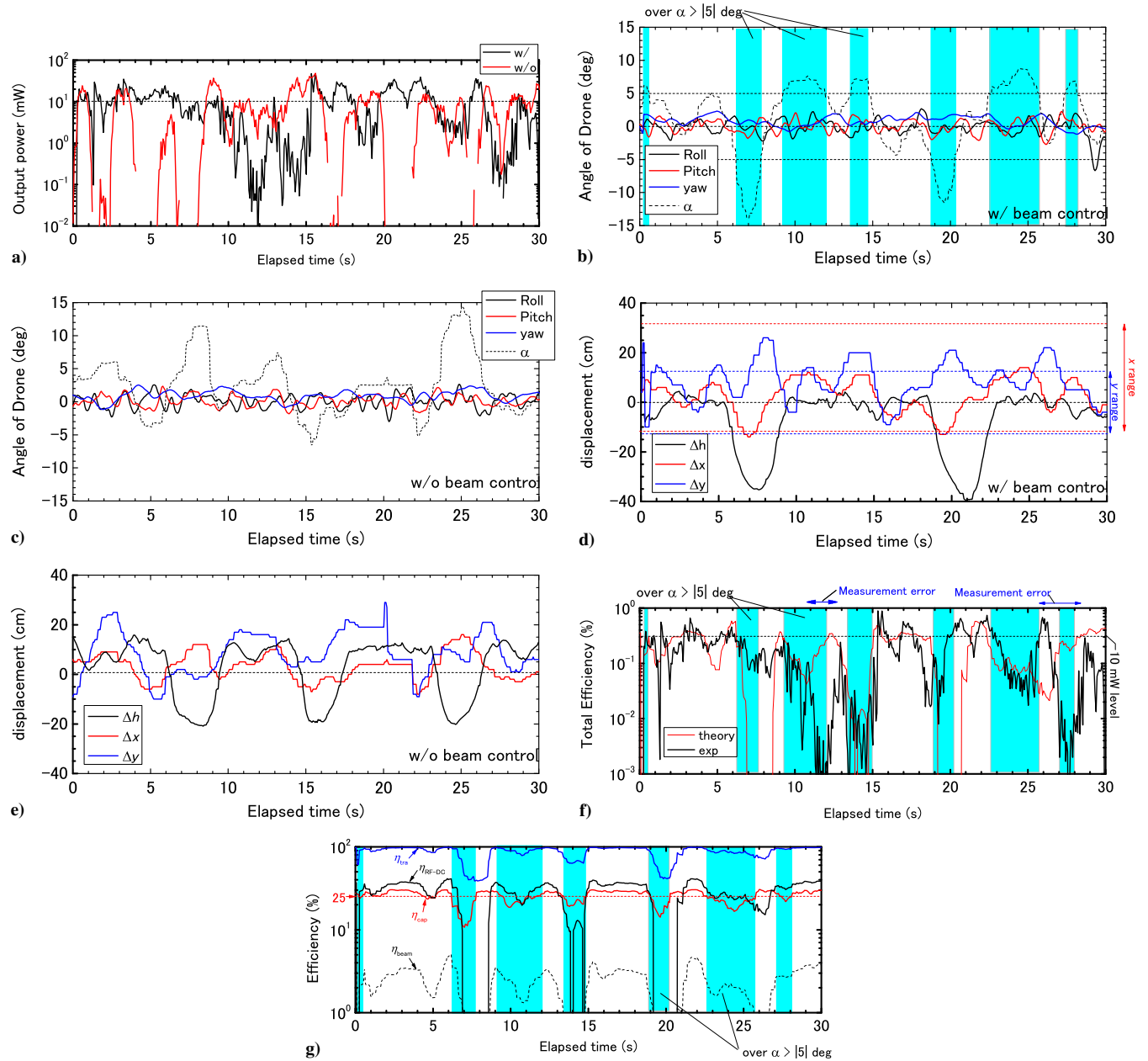


Fig. 4 Time variation of DC output power a), flight parameters b)–e), and WPT efficiency f),g) for the comparison with and without the beam control.

The peak efficiency of total WPT is slightly different in both cases. Subtracting the RF-DC rectifier efficiency to compare with the previous study of the rocket [3] gives a WPT between two antennas of about 2.3% in the present study. Further investigation is thus necessary to improve the WPT efficiency and increase the flight distance for the rocket-launching demonstration.

V. Conclusions

To demonstrate and evaluate WPT technology for aerospace craft and rockets, the microwave phase from the ground transmitter was controlled to compensate for location scattering of UAVs. During hovering flight at a height of 800 mm, two horn antennas were used to emit microwaves at the tracking angle $\alpha = \pm 10^\circ$. The resulting microwave transmission efficiency was about 10^{-2} for a 30 s flight, except for position errors due to the loss of communication. According to the analytical formula, η_{beam} , η_{cap} , η_{tra} , and $\eta_{\text{RF-DC}}$ are 4, 30, 90, and 40%, respectively, for a total transmission efficiency of 0.43%. The experiment described herein thus determines the WPT efficiency for a free-flight object. Improving the tracking system and increasing

the transmission distance is a subject for future investigation into beamed-energy propulsion.

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